

ME315 - Lab07

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SQLite - Parte II

Atividade

1. Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
library(RSQLite)

conn <- dbConnect(SQLite(), "database.sqlite3")
conn
```

```
<SQLiteConnection>
  Path: /cloud/project/database.sqlite3
  Extensions: TRUE
```

```
dbListTables(conn)
```

```
[1] "course_offerings"  "courses"           "grade_distributions"
[4] "instructors"       "rooms"              "schedules"
[7] "sections"          "subject_memberships" "subjects"
[10] "teachings"
```

```
dbListFields(conn, "instructors")
```

```
[1] "id"  "name"
```

```
dbListFields(conn, "teachings")
```

```
[1] "instructor_id" "section_uuid"
```

```
dbListFields(conn, "grade_distributions")
```

```
[1] "course_offering_uuid" "section_number"      "n_count"  
[4] "c_count"              "f_count"              "cr_count"  
[7] "p_count"              "nw_count"             "other_count"  
[10] "bc_count"             "u_count"              "d_count"  
[13] "ab_count"             "a_count"              "s_count"  
[16] "i_count"              "b_count"              "nr_count"
```

```
dbListFields(conn, "sections")
```

```
[1] "uuid"                  "course_offering_uuid" "section_type"  
[4] "number"                "room_uuid"           "schedule_uuid"
```

```
dbListFields(conn, "course_offerings")
```

```
[1] "uuid"          "course_uuid" "term_code"   "name"
```

```
dbListFields(conn, "courses")
```

```
[1] "uuid"  "name"  "number"
```

```
dbListFields(conn, "schedules")
```

```
[1] "uuid"      "start_time" "end_time"   "mon"      "tues"  
[6] "wed"       "thurs"      "fri"        "sat"      "sun"
```

```
dbListFields(conn, "subject_memberships")
```

```
[1] "subject_code"      "course_offering_uuid"
```

```
dbListFields(conn, "subjects")
```

```
[1] "code"          "name"          "abbreviation"
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT');

```
sql <- paste("SELECT DISTINCT instructors.id, instructors.name FROM instructors",
             "INNER JOIN teachings ON teachings.instructor_id = instructors.id",
             "INNER JOIN sections ON sections.uuid = teachings.section_uuid",
             "INNER JOIN subject_memberships",
             "  ON subject_memberships.course_offering_uuid = sections.course_offering_uuid",
             "INNER JOIN subjects ON subjects.code = subject_memberships.subject_code",
             "WHERE subjects.abbreviation = 'STAT'",
             "ORDER BY instructors.name")
profs <- dbGetQuery(conn, sql)
head(profs)
```

	id	name
1	4787911	ABIGAIL BENZINE
2	3111842	ADIN-CRISTIAN ANDREI
3	3775809	AKICHIKA OZEKI
4	4958796	ALAN HUANG
5	5451706	ALBERTO DEL PIA
6	3661782	ALBRECHT KLEMM

```
nrow(profs)
```

```
[1] 379
```

Resposta: A tabela apresenta o nome dos professores que lecionaram disciplinas de estatística, havendo 379 deles.

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):
- Quem é o professor mais difícil?
 - Quem é o professor mais fácil?

- Qual é a disciplina mais difícil?
- Qual é a disciplina mais fácil?

```
sql <- paste(
  "CREATE TEMP TABLE gpa_stat AS",
  "SELECT DISTINCT teachings.instructor_id AS instructor_id,",
  "course_offerings.course_uuid AS course_uuid,",
  "grade_distributions.course_offering_uuid AS offering,",
  "grade_distributions.section_number AS section_number,",
  "(4.0*a_count + 3.5*ab_count + 3.0*b_count + 2.5*bc_count +",
  "2.0*c_count + 1.0*d_count) /",
  "(a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) AS gpa,",
  "(a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) AS n",
  "FROM grade_distributions",
  "INNER JOIN sections",
  "ON sections.course_offering_uuid = grade_distributions.course_offering_uuid",
  "AND sections.number = grade_distributions.section_number",
  "INNER JOIN teachings ON teachings.section_uuid = sections.uuid",
  "INNER JOIN course_offerings",
  "ON course_offerings.uuid = grade_distributions.course_offering_uuid",
  "INNER JOIN subject_memberships",
  "ON subject_memberships.course_offering_uuid = course_offerings.uuid",
  "INNER JOIN subjects ON subjects.code = subject_memberships.subject_code",
  "WHERE subjects.abbreviation = 'STAT'",
  "AND (a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) > 0")
dbExecute(conn, sql)
```

[1] 0

```
sql_prof <- paste("SELECT instructors.name, AVG(gpa) AS gpa_medio, SUM(n) AS alunos",
  "FROM gpa_stat",
  "INNER JOIN instructors ON instructors.id = gpa_stat.instructor_id",
  "GROUP BY instructors.id, instructors.name",
  "ORDER BY gpa_medio")
dbGetQuery(conn, paste(sql_prof, "LIMIT 1")) # mais difícil
```

	name	gpa_medio	alunos
1	JAMES D KUELBS	2.6375	40

```
dbGetQuery(conn, paste(sql_prof, "DESC LIMIT 1")) # mais fácil
```

```

              name gpa_medio alunos
1 SRIKANTHMADHAVAN ARAVAMUTHAN      4      16

```

```
# Disciplina mais difícil / mais fácil
```

```

sql_disc <- paste("SELECT courses.name, courses.number, AVG(gpa) AS gpa_medio, SUM(n) AS alu
                  "FROM (SELECT DISTINCT course_uuid, offering, section_number, gpa, n",
                  "      FROM gpa_stat) AS g",
                  "INNER JOIN courses ON courses.uuid = g.course_uuid",
                  "GROUP BY courses.uuid, courses.name, courses.number",
                  "ORDER BY gpa_medio")

```

```
dbGetQuery(conn, paste(sql_disc, "LIMIT 1")) # mais difícil
```

```

              name number gpa_medio alunos
1 Introduction to the Theory of Probability 431 2.891568 2231

```

```
dbGetQuery(conn, paste(sql_disc, "DESC LIMIT 1")) # mais fácil
```

```

              name number gpa_medio alunos
1 Nonparametric Statistics and Machine Learning Methods 841      4      24

```

Resposta:

- JAMES D KUELBS é o professor mais difícil.
 - SRIKANTHMADHAVAN ARAVAMUTHAN é o professor mais fácil.
 - “Introduction to the Theory of Probability” é a matéria mais difícil.
 - “Nonparametric Statistics and Machine Learning Methods” é a matéria mais fácil.
4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

Data e Horário

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-23 22:05:09 -03"
```